

Collective deterrence as a classification problem: Voting rules, deterrence credibility, and escalation risk

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Abstract

Deterrence coalitions that collectively own their deterrence technology need an institutional design to decide when to retaliate against an attack or incident. This choice of institutional design, formalized through a social choice function, introduces a tradeoff between credible deterrence and escalation control. We study this tradeoff via a simple signalling model, and use it to construct an associated binary classification problem to determine institutional designs that perform well on a variety of environments. For a small coalition of four members, we compute and study the statistics of the empirical ROC curves associated to a variety of choice functions and probability distributions for retaliation and false positives.

Convince your enemy that he will gain very little by attacking you, which will diminish their enthusiasm.

– Sun Tzu, *The Art of War* [29].

1 Introduction

Since the emergence of modern warfare, understanding how to deter potential attackers has become a central concern of strategic studies and a core element of state security. Deterrence as a military strategy – avoiding an attack by maintaining sufficiently powerful means of guaranteed retaliation [24] – rests on shaping an adversary’s beliefs such that the expected costs outweigh the perceived benefits.

In this simplified form there are two strategies, reduce the potential benefits or increase the costs. The former is often counter-productive, though has been pursued through tactics such as scorched-earth during the second world war, see for example [21]. The latter, how-

ever, requires that the potential attacker believes the consequences to be large, for example by believing that large scale retaliation would occur.

In essence, deterrence relies on a push-pull between two fundamental objectives that are, in effect, the strategic analogues of true and false positive classifications [23]:

1. the resolve to use retaliatory means have to be sufficiently credible, and
2. one should limit the possibility of escalation spirals and accidental use.

The dynamic between these two mechanisms is especially potent for nuclear deterrence, where the consequences are potentially of an existential nature. For a classical outlook on these fundamental issues see [23] and [26]. Classical deterrence theory, however, do not often examine coalitions and their collective deterrence capabilities, which is the goal of this paper.

NATO is the only proper nuclear alliance – in the sense that of being a multinational alliance and not a bilateral agreements – where the deterrence capabilities of individual nations are also shared to allies. However, NATO does not own nuclear weapons as a collective entity, nor does it operate a jointly owned nuclear command structure with shared launch authority. For example, U.S. weapons stationed in Europe remain under exclusive U.S. custody and control, and any nuclear mission requires authorization from the relevant national leaders in addition to political approval through NATO’s Nuclear Planning Group (NPG).

Hence, there is currently no coalition of states in the world that jointly own both the nuclear deterrence capabilities and the command structure to act upon threats, as the historical proposals for a NATO multinational nuclear force never materialized [28]. However, such coalitions might become more politically viable now that arms-control frameworks, such as the New

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START treaty [30], are intensifying the European discussions about burden sharing and the possible future role of France or the UK in European nuclear deterrence.

Currently, of course, the Non-Proliferation Treaty ([31]) serves as a block to the creation of such nuclear alliances. However, there is room to break the treaty in the case of existential threats, or, alternatively, one could consider other deterrence technologies not limited by such restrictions. In the advent of such coalition then, the additional problem of *collective* rather than *unilateral* decision-making arises to complicate the above two problems. The problem of *credible resolve* becomes a problem of how the coalition’s states aggregate their individual resolves into a joint decision, and the problem of *escalation control* becomes a problem of not allowing this joint decision to be too easily affected by individual potential errors. Hence this joint institutional decision-structure introduces a trade-off between the two fundamental problems.

We can then phrase the motivating question for the present paper as: How does the institutional design of a coalition affect the deterrence mechanism and its effectiveness? In this paper we study this question as a classification problem using Receiver Operating Characteristic (ROC) curve analysis.

Aggregations of individual choices are in social choice theory measured by a *social choice function*, see for example [2, 19]. To set up our method, we develop a simple Bayesian decision model inspired by *signalling games* [12]. This model is dependent on a weighted threshold choice function f that aggregates the coalition’s votes on retaliation into a shared resolve.

To understand the interactions between deterrence and institutional design, we compute simple attack conditions, and use this to formulate the choice of institutional design for the coalition as a *binary classification problem*. Each institutional design has an associated empirical ROC curve – see [9] – for each information environment, formalized by vectors of probabilities of resolves and false positives for the coalition members. By studying the statistics of these ROC curves for a small hypothetical coalition of four members, we can study which designs that perform well and which do not. This does not capture the full strategic payoffs for the deterrence coalition, but gives a tractable and

computable way to compare social choice functions on their detection abilities.

A voting scheme that performs well on a variety of probability distributions for retaliation and false alarms is, perhaps not surprisingly, the unbiased one, where each coalition members has equal say. Close behind follows a scheme where each coalition members is given a weighted vote according to some relevant capabilities, like ISR. Across our stylized information environments, the worst performer is the dictatorial scheme, where only one country has voting power over the use of deterrence technology.

Finally we use Youden’s J -statistic to study which thresholds that give the best tradeoff between credible deterrence and escalation control, concluding that the best threshold usually is 2 for our small hypothetical coalition of four members.

Our goal with this paper is to provide a first step toward understanding the interaction between institutional design and collective deterrence by introducing, studying and exploring a novel binary classification-based framework. We deliberately adopt a simplified game model to isolate the informational tradeoffs inherent in collective decision-making – extensions incorporating more strategic behavior, correlated information, and proofs of optimality will be studied in future work.

1.1 Contributions

This paper has three main contributions towards the understanding of institutional design in collective deterrence:

1. introducing a novel classification-based framework for analyzing collective deterrence, with an easily understandable visual element;
2. providing a tractable model for linking institutional design to deterrence credibility and escalation risk;
3. illustrating the framework through simulations of small coalitions under varying informational environments.

Our results should be interpreted as illustrative patterns and not general results, as they are based on statistical analysis for a small coalition. The results,

however, give clear directions for future work on optimal institutional designs using a more general and proof-based approach.

2 Theoretical background

Our model exists as a combination of two much studied mathematical models: signalling games and social choice functions. These are each ubiquitous in their own respective fields of deterrence theory and voting theory. Before putting forth our combined model, we briefly explain the pieces separately.

2.1 Signalling games

Signalling games are dynamic Bayesian games, first studied in relation to learning by Lewis [16]. For a more modern treatment see [12], and for deterrence specific signalling games see [23].

In a signalling game the players have incomplete information about the initial situation, and the game unfolds sequentially over time, with actions known to the other players [12]. More precisely, one player – often called the sender – has some private information about the state of the world, and takes an observable action S , called the signal, after which another player – often called the receiver – updates their beliefs about the world and chooses an action. The private information of each player is called the *type*.

In our case of deterrence, incomplete information is essential; the uncertainties around the opponents resolve lies at the heart of the mechanism making deterrence a viable strategy.

In non-cooperative game theory, Nash equilibria are the standard solution concept for determining optimal strategies. In signalling games, however, one usually needs more refined concept, due to the players not knowing the beliefs of the other players correctly. These are called *perfect Bayesian equilibria*, see [6].

We will, however, not solve for this solution concept in this paper, as our game will be of a deliberately simple nature where this full generality is unnecessary. Instead we will use *attack conditions*, which are assignments of simple actions to the type of the players. These are based on incomplete-information thresholds

in crisis bargaining, see for example [10], and is a standard simple way of studying simple strategic actions.

2.2 Social choice functions

Social choice functions are mathematical models for ranking a set of alternatives, often called *social states*, by their collective desirability – combining the independent preferences of the individual into a common “will” of the people. Early versions were studied by Condorcet and Lull, see [8] and [7], but the modern theory now in use is often attributed to Arrow’s work [2] – see [19] for an introduction.

Social choice functions can be either *cardinal* or *ordinal*, [25]. Ordinal functions only use the relative ordering of social states to compute the best preferred state according to some mathematical rule. On the other hand, cardinal functions use specific values attached to each social state, and hence is able to compute not just which state is preferred, but how much it is preferred to the other states. It is the former, the ordinal choice functions, that are most often used in public voting and electoral systems, and also the type of function we will use in this paper. Our needs are also very simple, as our list of potential outcomes will be binary, hence we focus only on these in the following examples.

The most frequently used voting rule is the *majority rule*, stating the preferred choice between two options is the one chosen by more than half of the voters [17]. In mathematical terms,

$$f(v; w) = \begin{cases} 1, & \sum_{i=1}^n v_i > \frac{n}{2} \\ 0, & \text{else.} \end{cases}$$

By varying the threshold for the rule outputting a 1, we also vary the choice function. For example, the consensus rule, used for example when NATO votes on shared policies and solutions [20], states that the preferred choice between two options is the one chosen by *all* voters:

$$f(v; w) = \begin{cases} 1, & \sum_{i=1}^n v_i = n \\ 0, & \text{else.} \end{cases}$$

This is essentially a voting scheme in which every voter has a veto right. Choosing other thresholds, like $\frac{2}{3}$

gives the qualified majority rule and choosing $\frac{3}{4}$ gives the *supermajority rule*.

There are also many other voting schemes, like the *dictatorial* scheme, where only one voter has a say and always completely determines the outcome. The restricted class of social choice functions used in the model is described in [Section 3.2](#).

3 The model

Our model adopts this basic structure of a signalling game, as presented above, where we assume one of the players to be a coalition consisting of multiple states $\mathcal{C} = \{1, 2, \dots, n\}$. The other player is some potential attacking adversarial actor $\mathcal{A}$. Before we explain the individual parts, we summary the sequence of events:

1. the coalition has a predetermined fixed social choice function f , which is publicly known;
2. nature draws aversary type T ;
3. nature draws signals s_i for each coalition member;
4. adversary actor $\mathcal{A}$ chooses to attack or not based on the above information;
5. in the case of an attack, the coalition votes on retaliation using f .

3.1 Types and signals

The adversary $\mathcal{A}$ has a *type*, $T \in \{0, 1\}$, where 1 represents being aggressive – actively planning an attack – and 0 represents a benign adversary. More precisely, this is supposed to signify the intent to attack; if $T = 1$ then $\mathcal{A}$ intends to attack unless deterred, and if $T = 0$, then $\mathcal{A}$ prefers peace, but might still launch small probing attacks if the probability of retaliation is low enough. The coalition has a shared prior belief $\pi = \Pr(T = 1)$.

Each member i of the coalition $\mathcal{C}$ has a binary signal $s_i \in \{0, 1\}$, which is understood as a reduced-form decision-relevant indicator, summarizing both informational assessments about the adversary $\mathcal{A}$ and the member’s willingness to retaliate. This is dependent on their belief of the type of $\mathcal{A}$, giving a probability

$$p_i = \Pr(s_i = 1 \mid T = 1)$$

for whether coalition member i will vote to retaliate or not. Similarly, coalition members might receive false alarms, which gives a probability

$$q_i = \Pr(s_i = 1 \mid T = 0)$$

of a retaliatory strike based on erroneous information.

Remark 3.1. We assume here that signals and probabilities are independent across members, even though this is highly unlikely in a real scenario. In an actual coalition, information and intelligence is shared, discussions are held and decision-makers are influenced by other members. This said, the choice allows us to gain computational power and simplicity. With our focus on understanding the underlying *mechanisms* and not be as true-to-reality as possible, we feel that this simplification is sufficiently justified.

In the case of an attack, each coalition member $i \in \mathcal{C}$ then decides on a binary vote $v_i \in \{0, 1\}$, interpreted as their vote towards authorizing retaliation. In order to focus this paper on the institutional mechanisms, and not on whether the coalition members try to strategize for their own gain, we assume that $v_i = s_i$, which means that we assume the actors to vote truthfully based on their own intelligence assessments and internal stance.

3.2 Weighted threshold functions

The coalition is assumed to have a publicly known institutional design, specifically a decision rule formalized by a social choice function f . In order to be able to analyze the behaviour of our signalling game with respect to such a function, we need to restrict to a class of viable rules. First, we assume a binary outcome, 1 representing coalition retaliation and 0 representing no retaliation. Second, we assume the choice function f to be a weighted threshold rule. This means that each coalition member i is assigned a weight $w_i \in \mathbb{R}$ dependent on the context of the attack, and that there is a threshold $\tau \in \mathbb{R}$, such that the aggregated vote of the coalition is 1 if the weighted sum of the individual members votes exceeds the threshold. This weight scheme is meant to include the possibility that the attacked country in the coalition gets a vote that counts more towards the total, or that frontline countries have a larger say, or that a country has far better ISR capabilities etc.

Formally, then, given a list of votes coalition individual votes $v = (v_1, v_2, \dots, v_n)$, and a weight scheme $w = (w_1, \dots, w_n)$, we have

$$f(v; w) = \begin{cases} 1, & \sum_{i=1}^n w_i v_i \geq \tau \\ 0, & \text{else} \end{cases}$$

Most viable voting rules are formalizable in this framework, like the majority rule, consensus rule, dictatorship, qualified majority, veto players, etc. There are also some decision rules that are not possible to model in this framework, like multi-stage rules, lexicographical rules, non-monotonic rules and randomized rules. However, such rules are very unlikely to be viable for choosing a retaliation action, hence our focus on weighted threshold rules.

3.3 Retaliation probabilities

Assuming $\mathcal{A}$ attacks, retaliation will occur if the weighted number of coalition members receiving the signal meets the threshold. We denote the probability of retaliation by

$$R(\tau) = \Pr \left(\sum_{i=1}^n w_i v_i \geq \tau \mid T = 1 \right)$$

This function captures the *deterrence credibility* of the coalition $\mathcal{C}$. In other words it is the probability that an attack by the adversary $\mathcal{A}$ triggers a retaliatory response.

Similarly, if the adversary does *not* attack, then retaliation is accidental, with probability

$$F(\tau) = \Pr \left(\sum_{i=1}^n w_i v_i \geq \tau \mid T = 0 \right)$$

This function captures the coalition's *escalation risk*, i.e., the probability that the coalition accidentally authorizes a response. This could happen if, for example, another actor than $\mathcal{A}$ was responsible for an attack on the coalition, or that an accident is interpreted as an attack, as explored in multiple wargames and academic studies, see for example [22] and [4].

Example 3.2. In the simple case that each coalition member i has an equal weight, say $w_i = 1$ for all i , as well as equal probabilities p_i , and the threshold τ is

a positive integer less than or equal to n , then these probabilities become binomial, with

$$R(\tau) = \sum_{k=\tau}^n \binom{n}{k} p^k (1-p)^{n-k},$$

and similarly

$$F(\tau) = \sum_{k=\tau}^n \binom{n}{k} q^k (1-q)^{n-k}.$$

The often used majority rules and consensus rules are both of this nature.

3.4 Costs and benefits

The preferred outcome of the adversary actor $\mathcal{A}$ is to attack and suffer no consequences. The preferred outcome for the coalition $\mathcal{C}$ is that no attack occurs.

In order for an attack to be of interest, there has to be a perceived benefit for $\mathcal{A}$. In the situation of preferred outcome for an adversary $\mathcal{A}$ of aggressive type $T = 1$, where it attacks and no retaliation occurs, we denote this benefit by B_1 . We assume this value to be a number, for example monetary benefit or some other measure of benefit that $\mathcal{A}$ deems desirable. If retaliation occurs, then $\mathcal{A}$ suffers a cost C_1 , which we assume is of comparable type to the benefit B_1 .

Similarly, if the adversary has benign type $T = 0$, then there might still be a small benefit B_0 for initiating an attack, if the probability of retaliation is small enough. This tries to capture low-stakes provocations and probing actions. Similarly, there is a cost C_0 . These are assumed to be smaller than the payoffs for the aggressive type, giving inequalities

$$B_0 < B_1 \text{ and } C_0 < C_1.$$

We can then compute the expected payoffs for an adversary of a given type. The probability of retaliation by $\mathcal{C}$ is given as above by $R(\tau)$, hence the expected payoff for $\mathcal{A}$ with type $T = 1$ for attacking is

$$E_{T=1}(\text{attack}) = (1 - R(\tau))B_1 - R(\tau)C_1,$$

and similarly

$$E_{T=0}(\text{attack}) = (1 - R(\tau))B_0 - R(\tau)C_0.$$

3.5 Attack conditions

As our model is very simple from a game-theoretical perspective, the attack conditions are quite easy to find. In our model, only $\mathcal{A}$ is a strategic actor, who makes a choice of an action based on some information – the coalition $\mathcal{C}$ does not perform a strategic action, it is decided upon by their predetermined institutional design.

Assuming rationality, the adversary state attacks if it deems the expected payoff to be greater than 0. Rearranging the terms gives that $\mathcal{A}$ of type $T = 1$ attacks if and only if

$$R(\tau) < \frac{B_1}{B_1 + C_1},$$

and that $\mathcal{A}$ of type $T = 0$ attacks if and only if

$$R(\tau) < \frac{B_0}{B_0 + C_0}.$$

Remark 3.3. This setup deliberately collapses the coalition’s response into a single probabilistic parameter. This is done to later allow us to isolate the contribution from the institutional design to this parameter.

These fractions give the adversary actor $\mathcal{A}$ ’s *deterrence thresholds* and *deterrence intervals*. For simplicity, denote by $\bar{R}_1 := \frac{B_1}{B_1 + C_1}$ and $\bar{R}_0 = \frac{B_0}{B_0 + C_0}$. By the inequalities above, we get $\bar{R}_0 < \bar{R}_1$. This means we have three possibilities, which corresponds to the three possible regions:

1. if $R(\tau) < \bar{R}_0$, then the adversary actor $\mathcal{A}$ attacks, no matter its type;
2. if $\bar{R}_0 < R(\tau) < \bar{R}_1$, then a benign adversary is deterred, but an aggressive one attacks;
3. if $R(\tau) > \bar{R}_1$, then the coalition $\mathcal{C}$ ’s collective deterrence has succeeded in deterring also an aggressive adversary.

Remark 3.4. We will not use this in any degree, but these three strategies and beliefs constitute the perfect Bayesian equilibria for this game. The first and last regions above give pooling equilibria, while the second is a *separating* equilibrium. They are also the only possible ones, and thus yields a full classification.

Remark 3.5. The costs C_1 and C_0 depends on the coalition $\mathcal{C}$ ’s available deterrence technology. For exam-

ple, having strategic nuclear weapons significantly increases the costs of retaliation for the adversary $\mathcal{A}$. As discussed in the introduction, new nuclear alliances are prohibited by the Non-Proliferation Treaty [31]. But, in certain situations and contexts, it is possible to circumvent this, or alternatively use other available or future technologies. We can then study two different costs – one with and one without sufficient deterrence technologies – and see how this affects the equilibria. Doing this reveals the natural outcome: larger costs means smaller expected payoffs, and larger intervals where deterrence is successful.

As the game is very simple, it is not only the attack conditions that are interesting. The interesting question about our setup is, in our opinion, trying to determine which social choice functions that yield the best deterrence properties for the coalition $\mathcal{C}$. In the next section we attempt to analyze this question. Our method of choice does not utilize the full strategic information available, but is based purely on the probabilities of “true” and “false” retaliations. This is to isolate the effect of the institutional design on these probabilities as a first step towards a broader understanding of the full dynamics.

4 Good schemes and thresholds

For a given coalition $\mathcal{C}$, there are countless possible choices for an institutional design f that computes the aggregated decision based on coalition members’ votes. As mentioned earlier, this function f always introduces a tradeoff between credible deterrence and escalation control.

For $\mathcal{C}$ then, it is highly important to understand which such designs that give the best deterrence credibility while minimizing the risk of accidental retaliation. In essence, this is the classical tradeoff between the probability of *true positives* and the probability of *false positives*. There is a rich history from many fields of science studying this exact tradeoff, see for example [13] and [9], that we can use to study this issue in our setting.

Our choice function f has two possible outcomes, retaliation or no retaliation, based on two possible situations of true or false attacks. By false attack, or no attack, we here mean that there has been a false detection due to an accident, or an attack that is misattributed to

the wrong adversary. This means that the function f is a binary classifier, in the sense of [15].

The standard way of measuring the accuracy of a binary classifier is to measure the correct and incorrect assignments, [1]. For this we have a simple contingency matrix:

	Retaliation	No retaliation
Attack	True positive	False negative
No attack	False positive	True negative

This gives us the *true positive rate* (TPR), defined as

$$\text{TPR} = \frac{\text{true positive}}{\text{true positive} + \text{false negative}}$$

and the *false positive rate* (FPR), defined as

$$\text{FPR} = \frac{\text{false positive}}{\text{false positive} + \text{true negative}}.$$

These are, respectively, the probability that retaliation occurs, and the probability of an accidental retaliation, which we have previously named $R(\tau)$ and $F(\tau)$.

The binary classifier is dependent on the threshold τ , which allows us to visualize the classifiers performance on the true positive-false positive tradeoff by ROC curves.

4.1 ROC curves

Receiver Operating Characteristic (ROC) curves were first used for radar signals during World War II, where they were used as an analysis tool for strengthening the predicting power of radars after the Pearl Harbor attack [32, 27]. Since then they have found a wide use throughout science, mainly in medicine and epidemiology [14], and more recently as a tool for evaluating classification algorithms in machine learning and artificial intelligence [3, 5].

The ROC space is defined by having TPR and FPR as its axes, and as these are rates from 0 to 1 the space is equivalent to the unit square. Any point in this space represents a tradeoff between true positives and false positives. The point $(0, 1)$ is the *perfect classifier*, in the sense that it correctly identifies 100%, with 0% errors. The diagonal line (x, x) represents a random classifier. Hence, points above the diagonal are better than random guessing, and points below are worse.

By plotting $(F(\tau), R(\tau))$ for all different thresholds τ , one gets a curve through ROC space that visualizes the effect of the threshold on the true positive-false positive tradeoff. The point on this curve that is closest to the perfect classifier $(0, 1)$ is the optimal threshold for the given choice function.

By making ROC curves for a variety of choice functions, we can then identify institutional designs for the coalition $\mathcal{C}$ that performs well in a variety of settings. One method for doing this is comparing ROC curves on their area under the curve (AUC) – a better institutional design will have a larger AUC. The AUC measures the probability that the given voting scheme ranks a true positive above a false positive [14], or in our case that it is able to clearly separate attacks from a known actor from “false alarms”. Hence, a large AUC – one that is close to 1 – will be more likely to produce large weighted sums when there is an actual attack, and low weighted sums when there is not.

After finding voting schemes that perform well across all thresholds, we study which threshold that gives the best tradeoff between credible deterrence and escalation control. We do this by comparing all thresholds on Youden’s J statistic, introduced originally in [33]. It is defined by $J(\tau) = R(\tau) - F(\tau)$, and is widely used in modern analyses, see for example [11]. The threshold with the largest J -number has maximal distance from the diagonal in the ROC space, and is interpreted as having the overall best detection quality.

We do this in detail for an example of a small coalition. The small number of members allows us to compute the ROC curves empirically for a variety of choice functions f . One can also easily do this for large coalitions by using normal approximation to the probabilities, and use binormal ROC curves [18]. We have, in light of future work, chosen to focus on a small coalition resembling the Nordic countries, although we will keep all detail hypothetical for this paper.

4.2 The needed computations

The votes $v_i \in \{0, 1\}$ are assumed to be independent, with probabilities p_i and q_i for the two types $T = \{0, 1\}$. Hence, the probability of a given vote-vector v is a

product of independent Bernoulli variables,

$$\Pr(v \mid T = 1) = \prod_{i=1}^n p_i^{v_i} (1 - p_i)^{1-v_i},$$

for the aggressive type $T = 1$, and similarly

$$\Pr(v \mid T = 0) = \prod_{i=1}^n q_i^{v_i} (1 - q_i)^{1-v_i}$$

for the benign type $T = 0$.

Given a weight vector $w = (w_1, \dots, w_n)$, we have for each vote-vector v a weighted sum $S = \sum_i^n w_i v_i$. There are 2^n possible vote-vectors, hence the distribution of this sum can be computed by aggregating all of the probabilities for all vote-vectors, with probabilities as above, producing two functions

$$\Pr(S = s \mid T = 1), \quad \Pr(S = s \mid T = 0).$$

Computing these probabilities allows us to also compute the points $(F(\tau), R(\tau))$ for all values of τ .

The computations in this paper were done with a simple python script, which can be downloaded from the authors GitHub page.

4.3 The example

Our small coalition $\mathcal{C}$ consist of four members. We will not go into specifics or create a realistic coalition here, as this paper only serves to demonstrate the effect of the institutional design on the deterrence mechanism, and not imply policy. This also allow us to test hypothetical alterations of the choice functions due to different contexts the coalition exists in; specifically we consider choice function weight vectors w based on hypothetical geographical, technological and geopolitical contexts.

In order to compare choice functions based on their performance, we plot ROC-curves for several natural weight schemes at the same time. We have chosen five different weight schemes for our coalition, which we will use for all of the probability variations. We have normalized them so that they all have the same total sum of 4, as well as put the weights in decreasing order for simplicity.

1. $w = (1, 1, 1, 1)$ – the *unbiased* scheme. This treats all coalition members equally. Varying the threshold τ then gives the standard voting schemes described in [Section 2.2](#).
2. $w = (4, 0, 0, 0)$ – the *dictator* scheme. This is essentially the shared capability structure employed by NATO today, where the country owning the deterrence technology also has full control over its use.
3. $w = (2.5, 0.5, 0.5, 0.5)$ – the *veto* scheme. The scheme allows a single member to have full veto rights for most thresholds.
4. $w = (1.16, 1.06, 0.89, 0.89)$ – the *technology* scheme. This weighs members according to bias coming from asymmetric qualities, like techonology, competencies, or ISR capabilities.
5. $w = (1.6, 0.8, 0.8, 0.8)$ – the *frontline* scheme. This scheme weighs, for example, a coalition member that is a border-state with the adversary higher than the other members.
6. $w = (1.6, 1.2, 0.8, 0.4)$ – the *geographical* scheme. The scheme weighs members according to their distance from the aggressor, and hence by the probability of being affected.
7. $w = (1.33, 1.33, 0.67, 0.67)$ – the *two-bloc* scheme. This represents a coalition where there are two main members forming a block, and two secondary members forming a second block.

To have a base line, we look at the case where all coalition members have the same probability of voting in favor of retaliation given the type of $\mathcal{A}$. [Fig. 1](#) shows the ROC curves for the above weight schemes when both the shared resolve and the accuracy for correct identification are *high* – represented by $p = [0.85, 0.85, 0.85, 0.85]$ and $q = [0.05, 0.05, 0.05, 0.05]$.

As is perhaps not too surprising, all choice functions perform fairly well when the probabilities are very simple to distinguish. All AUCs are close to 1, meaning that the probability of ranking an attack above a false alarm is almost guaranteed for all of the schemes. The unbiased scheme, the technology scheme and the two-bloc scheme performs the best, with an AUC of 0.998, while the dictator scheme performs the worst, with an AUC of 0.879, which is still not too bad.

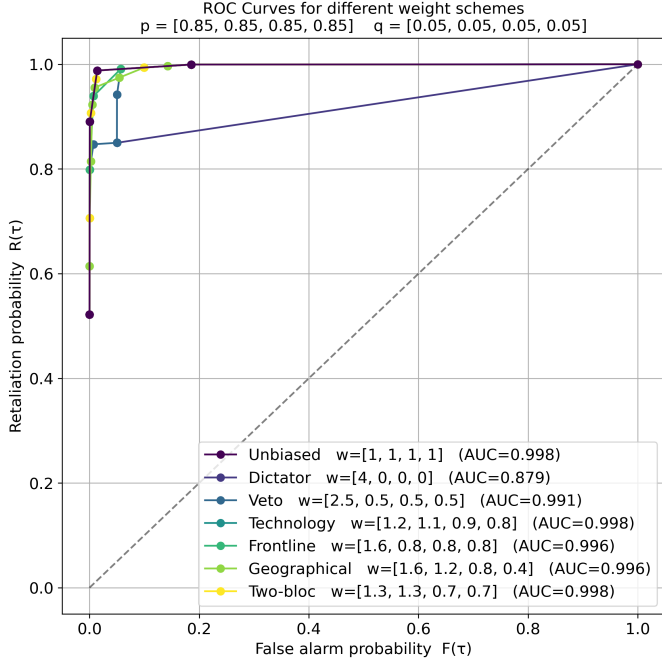


Figure 1: ROC curves for high accuracy and resolve.

Similarly, we consider in Fig. 2 the case when the shared resolve and accuracy is *low* – represented by $p = [0.60, 0.60, 0.60, 0.60]$ and $q = [0.20, 0.20, 0.20, 0.20]$.

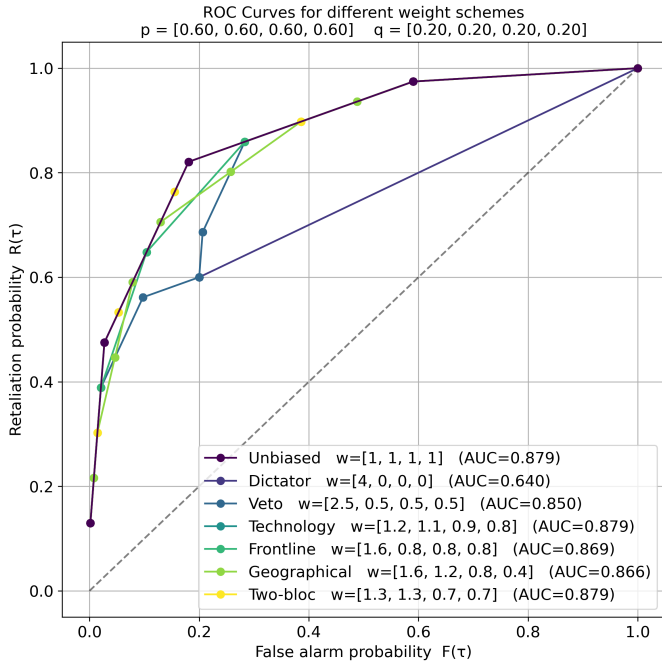


Figure 2: ROC curves for low accuracy and resolve.

As each probability is still equal for all members, the shape of the ROC curves are identical to the ones above, only pushed in closer to the diagonal. Most of the voting schemes still perform fairly well given the circumstances.

To differentiate a bit more, we consider the following decreasing probability-vector $p = [0.85, 0.70, 0.55, 0.40]$, with the corresponding increasing vector $q = [0.05, 0.10, 0.20, 0.30]$. These probabilities align with the ranking of the members in all the weight schemes that are not the unbiased one, meaning that they “match” the hypothetical scenario. We then get the ROC curves in Fig. 3.

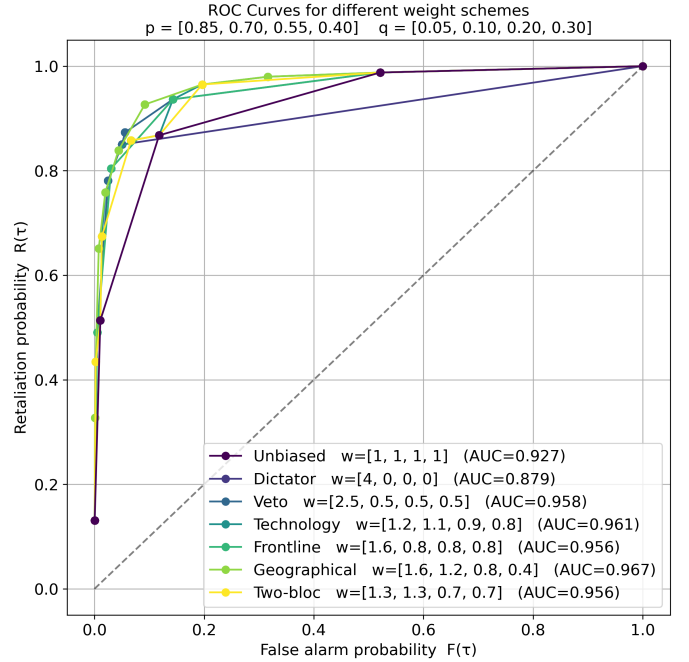


Figure 3: ROC curves for a decreasing probability vector.

Here we see that the voting schemes have different performances, and that the unbiased scheme is no longer the top performer. The best performing scheme is the one that best utilizes the distribution of the probabilities – the geographical scheme. The dictator scheme still performs the worst.

We also consider the case when the probabilities are not oriented in the same direction as the weights, using the “misaligned” vectors $p = [0.50, 0.60, 0.75, 0.90]$ and $q = [0.40, 0.30, 0.15, 0.05]$, see Fig. 4

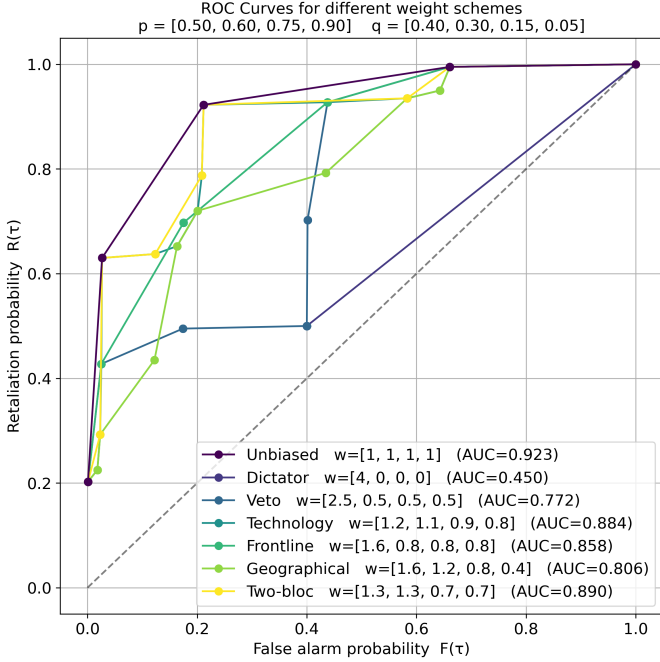


Figure 4: ROC curves for an increasing probability vector.

This weight-probability combination exaggerates differences in voting schemes, which is also noticeable in the shape of the ROC curves – they are considerably more varied. The best performer is again the unbiased scheme. The worst performer is still the dictator scheme, which here has an AUC of under 0.5, meaning it is essentially worse than random guessing when trying to distinguish attacks from false alarms.

4.4 Scheme performance

As one can see, some of the weight schemes are performing consistently better than others. To further study this we evaluate the weight schemes on a large variety of probability vectors, listed below, and study the statistics of their AUCs. These form varying *information environments* that the coalition operates in. We use the probabilities as listed in Table 1.

Computing AUC values for all of these, we obtain the statistics for each weight scheme as in Table 2.

The best performer is the unbiased weight scheme. It has the highest mean AUC, as well as the highest minimum AUC, and the highest maximum AUC. It is, however, followed closely by the technology scheme and

p vector	q vector
(0.85, 0.85, 0.85, 0.85)	(0.05, 0.05, 0.05, 0.05)
(0.75, 0.75, 0.75, 0.75)	(0.10, 0.10, 0.10, 0.10)
(0.60, 0.60, 0.60, 0.60)	(0.20, 0.20, 0.20, 0.20)
(0.90, 0.80, 0.70, 0.60)	(0.05, 0.10, 0.15, 0.20)
(0.85, 0.70, 0.55, 0.40)	(0.05, 0.10, 0.20, 0.30)
(0.60, 0.65, 0.70, 0.75)	(0.25, 0.20, 0.15, 0.10)
(0.55, 0.65, 0.75, 0.85)	(0.35, 0.25, 0.15, 0.05)
(0.50, 0.60, 0.75, 0.90)	(0.40, 0.30, 0.15, 0.05)
(0.55, 0.55, 0.55, 0.55)	(0.45, 0.45, 0.45, 0.45)
(0.85, 0.55, 0.80, 0.50)	(0.05, 0.25, 0.10, 0.30)
(0.90, 0.88, 0.60, 0.50)	(0.05, 0.06, 0.25, 0.30)
(0.72, 0.58, 0.81, 0.63)	(0.11, 0.22, 0.07, 0.19)
(0.78, 0.82, 0.79, 0.40)	(0.12, 0.10, 0.11, 0.35)
(0.88, 0.55, 0.70, 0.83)	(0.06, 0.28, 0.18, 0.10)

Table 1: Varied range of vectors used in the analysis.

Scheme	Mean	Min	Max
Unbiased	0.932	0.606	0.998
Dictator	0.719	0.426	0.903
Veto	0.902	0.594	0.991
Technology	0.931	0.606	0.998
Frontline	0.922	0.602	0.996
Geographical	0.914	0.600	0.988
Two-bloc	0.929	0.606	0.998

Table 2: Mean, minimum and maximum AUC values for each weight scheme, computed from the probability-vectors listed in Table 1.

the two-bloc scheme, which shares its minimum and maximum. The dictator scheme performs considerably worse than the others, with the lowest mean, the lowest minimum and the lowest maximum AUC.

To visualize the performance of the top performer, we map all of its ROC curves against all the probability vectors in Table 1 in Fig. 5.

We see that its shape is consistent, and that its performance is good for most of the probability vectors. The curve close to the diagonal is associated to the vectors $p = [0.55, 0.55, 0.55, 0.55]$ and $q = [0.45, 0.45, 0.45, 0.45]$, which is close to being a random guess for the type of the adversary $\mathcal{A}$.

Similarly, for the technology scheme we get the collection of ROC curves presented in Fig. 6.

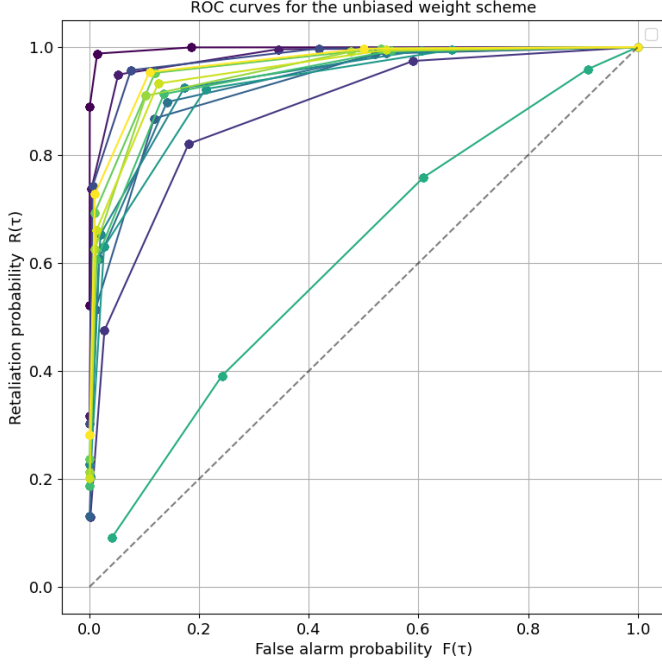


Figure 5: ROC curves for the unbiased scheme on a variety of information environments.

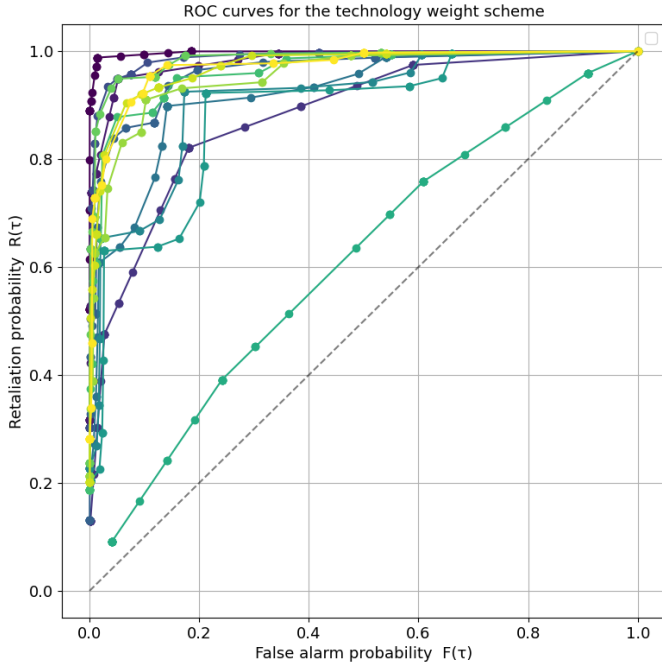


Figure 6: ROC curves for the technology scheme on a variety of information environments.

Its performance is statistically similar to the unbiased

scheme, but we see that it is much more sensitive to threshold variations, and that the resulting ROC curves are then more varied in shape. These alternative shapes come from probability vectors that are misaligned with the weight gradient – which is purposefully not representing the schemes power in utilizing individual differences. This would be like a member getting higher voting power for having better ISR technology, but then being consistently unable to utilize this technology to its full potential, while other members with worse technology manage it better.

Finally, Fig. 7 shows the same data for the worst performer – the dictator scheme.

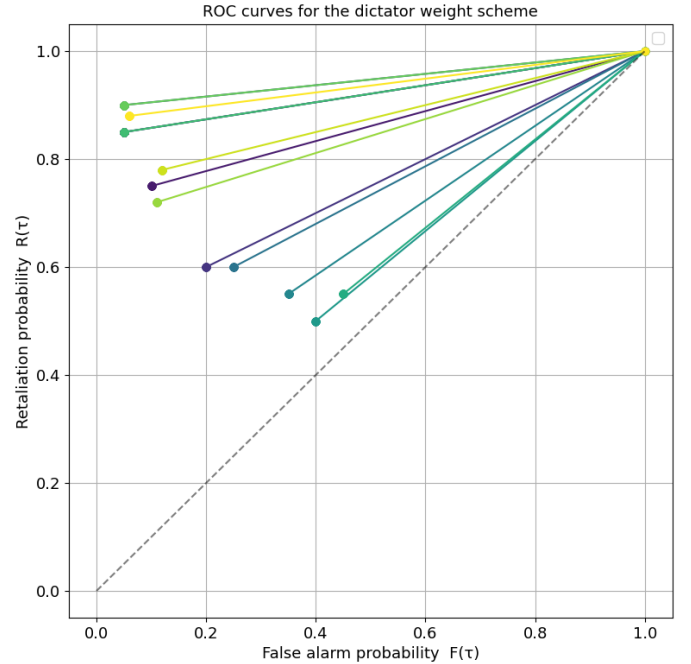


Figure 7: ROC curves for the dictator scheme on a variety of information environments.

Its ROC curves are all shaped like lines, as it is only possible for the scheme to output a 1 when the threshold is lower than the voting power of the dictator – all other thresholds give the same point (1,1). Note that the dictator scheme can perform well compared to the other schemes, in situations where the dictator member is vastly superior in information.

4.5 Threshold performance

The previous section argued that the voting scheme that performs , when ranked on their performance across *all* thresholds, is the unbiased scheme. This is, however, perhaps not the optimal metric for finding the *best* voting scheme for a small coalition. We then need to find the weight-threshold combination that gives the best performing single point – not the entire curve.

To do this we compute Youden’s J -statistic for all schemes on all of the probabilities in Table 1, and study the statistics of the schemes performances. Youden’s J -statistic is, as previously mentioned, defined by $J(\tau) = R(\tau) - F(\tau)$.

Just as for the AUC values, we list in Table 3 the mean, minimum and maximum J -numbers for all of the voting schemes, computed for all of the probability vectors in Table 1. We also list the mean optimal threshold τ^* , computed for the same data. The τ^* value below should really be discretized by rounding up to the nearest whole integer, but we have decided to keep them as is for comparison and insight.

Scheme	Mean	Min	Max	Mean τ^*
Unbiased	0.807	0.150	0.974	1.17
Dictator	0.566	0.100	0.850	0.10
Veto	0.705	0.125	0.934	0.84
Technology	0.802	0.150	0.974	1.44
Frontline	0.757	0.149	0.934	1.19
Geographical	0.760	0.149	0.946	1.48
Two-bloc	0.807	0.150	0.974	1.47

Table 3: The mean, minimum and maximum Youden’s J statistic, and mean optimal threshold τ^* for all voting schemes, computed across all probability vectors.

Here we see that the unbiased scheme and the two-bloc scheme performs the best overall, followed closely again by the technology scheme. The two-bloc scheme has a higher mean optimal threshold τ^* compared to the unbiased scheme, but in effect, due to the discrete small coalition, the best threshold is $\tau^* = 2$ for both.

As the unbiased scheme does not make any assumptions on the coalitions internal structure – the partitioning into blocs – as well as having better AUC statistics, we conclude that the better institutional design for the four-member deterrence coalition $\mathcal{C}$ is the *majority rule*.

Intuitively, the majority rule benefits from its essential feature: symmetry. When a member’s individual errors are independent and not systematically biased, the unbiased scheme’s equal weighting minimizes sensitivity to these miscalibrations, making it perform best on average.

Remark 4.1. If the coalition is highly certain and confident of their members technological capabilities and its accuracy in a plethora of real-world scenarios, then the technology scheme might outperform it – still with optimal threshold $\tau^* = 2$. However, given any possibility for a scenario where the probabilities are misaligned with the weights, the unbiased scheme will outperform the technology scheme on average.

5 Conclusion

Now that we have landed on a well-performing social choice function for our coalition $\mathcal{C}$, we can also understand the associated deterrence behavior for the signaling game in Section 3.

If we omit the unlikely scenario that the probability vectors are almost random, $p = [0.55, 0.55, 0.55, 0.55]$ and $q = [0.45, 0.45, 0.45, 0.45]$, we obtain an average retaliation probability

$$\overline{R(\tau^*)} = \frac{1}{13} \sum_{k=1}^{13} \Pr \left(\sum_{i=1}^4 v_i \geq 2 \mid T = 1, p = p_k, q = q_k \right) \approx 0.92$$

where p and q ranges over the remaining 13 probability vectors in Table 1. Similarly,

$$\overline{F(\tau^*)} = \frac{1}{13} \sum_{k=1}^{13} \Pr \left(\sum_{i=1}^4 v_i \geq 2 \mid T = 0, p = p_k, q = q_k \right) \approx 0.12.$$

On average, across a plethora of scenarios, the adversary state $\mathcal{A}$ attacks when the perceived benefit and costs satisfy

$$0.92 \leq \frac{B_1}{B_1 + C_1},$$

which happens if the costs are small compared to the benefits. Hence, for four-member coalitions with sufficient deterrence technologies, like strategic nuclear warheads, deterrence will be quite credible in most situations when using the majority rule as their institutional design.

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